ClinicaFlow

AI Diagnostic Summary Report

Patient Overview

• **Patient Name:** Ashley Grahm
• **Age:** 20
• **Gender:** Female
• **Primary Complaint:** Headache and rash

Red-Flag Alert

• SpO_2 = 80 % (significant hypoxia)
• Temperature = 100.4 °F (fever)
• Leukocytosis (WBC = 10,570 cells/µL)
• Potential early sepsis (hypoxia + fever + leukocytosis)

Risk Stratification

• **Level:** Critical
• **Key Drivers:**
• Fever (100.4 °F)
• Marked hypoxia (SpO_2 80 %)
• Elevated WBC count (10,570)
• New pruritic maculopapular rash
• Hyperglycemia & HbA1c 7.1 % (undocumented diabetes)
• Low vitamin B12 and high homocysteine
• High IgE (492 kU/L) indicating atopic tendency

Probable Diagnosis

• **Condition:** Community■acquired pneumonia with associated fever, headache and an acute pruritic maculopapular rash (possible viral co■infection or drug■related eruption). Underlying uncontrolled diabetes mellitus and vitamin B12 deficiency are contributing comorbidities.

• **Confidence:** 55 %

• **Justification (Explainability Pack):**

• The patient's fever, severe headache, and diffuse itchy rash together with hypoxia point toward a pulmonary infection complicated by a dermatologic manifestation.

• Laboratory data show leukocytosis, hyperglycemia, and an elevated HbA1c, supporting infection in the setting of diabetes.

• High IgE aligns with an allergic/atopic component to the rash.

- Absence of meningitic signs (neck stiffness, altered mental status) makes meningitis less likely.
- **Supporting Evidence:**
- Temperature 100.4 °F (fever flag)
- SpO_2 80 % (hypoxia flag)
- WBC 10,570 cells/µL (leukocytosis)
- New pruritic rash on arms and legs
- Elevated fasting glucose (141 mg/dL) and HbA1c (7.1 %)
- Low vitamin B12 (<148 pg/mL) and high homocysteine (23.86 µmol/L)
- High IgE (492 kU/L)

Differential Diagnoses

- **Acute allergic urticaria/atopic dermatitis flare** – High IgE and itchy rash; fever & leukocytosis are atypical.
- **Viral exanthem (e.g., measles, parvovirus, COVID■19)** – Fever, headache, rash, and possible respiratory involvement; COVID■19 can cause hypoxia.
- **Meningitis (bacterial or viral)** – Headache and fever present, but no neck stiffness, photophobia, or petechial rash.
- **Sepsis secondary to occult infection (e.g., urinary tract, skin infection)** – Leukocytosis, fever, hypoxia; however, vital signs lack tachycardia/hypotension.
- **Drug reaction (e.g., vitamin B complex hypersensitivity)** – Temporal relation to vitamin B supplementation; rash is itchy, yet systemic signs suggest infection.

Key Laboratory Findings

- **Leukocytosis:** WBC 10,570 cells/µL (high)
- **Hyperglycemia:** Fasting glucose 141 mg/dL (high)
- **Diabetes:** HbA1c 7.1 % (high)
- **Vitamin B12 deficiency:** <148 pg/mL (low)
- **Elevated homocysteine:** 23.86 µmol/L (high)
- **Elevated IgE:** 492 kU/L (high) – suggests atopic/allergic tendency
- **Vitamin D deficiency:** 25■OH Vitamin D 8.98 ng/mL (low)

Required Investigations

- **CBC with differential** – Confirm leukocytosis, assess anemia or neutrophil shift.
- **Comprehensive metabolic panel** – Evaluate electrolytes, renal & hepatic function.
- **Blood cultures ×2** – Rule out bacteremia/sepsis.
- **Chest X■ray** – Initial imaging to identify pneumonia or other pulmonary pathology.
- **Arterial blood gas** – Quantify severity of hypoxemia and acid■base status.
- **SARS■CoV■2 PCR** – Detect COVID■19 infection, which can cause rash and hypoxia.
- **Respiratory viral panel** – Screen for influenza, RSV, adenovirus, etc.
- **CT chest with contrast** – If X■ray is inconclusive or to evaluate for pulmonary embolism.
- **D■dimer** – Screen for PE if clinical suspicion arises.
- **Lumbar puncture** – Reserved for evolving neurological signs suggestive of meningitis.

- **Serum vitamin B12 repeat & methylmalonic acid** – Confirm deficiency severity.
- **Serum 25■OH vitamin D repeat** – Monitor response to supplementation.
- **HbA1c repeat in 3 months** – Ongoing diabetes monitoring.
- **Allergy skin testing or specific IgE panel** – Characterize atopic sensitization.
- **Skin biopsy of rash** – If rash is atypical or does not improve with antihistamines.

Suggested Medications / Treatments

- **Supplemental oxygen** – Titrate to maintain SpO_2 > 94 % (nasal cannula, mask, or high■flow as needed).
- **Empiric broad■spectrum antibiotics** for community■acquired pneumonia (e.g., amoxicillin■clavulanate or a macrolide) pending cultures.
- **Oral antihistamine** – Cetirizine 10 mg daily for itch relief.
- **Topical low■potency corticosteroid** – Hydrocortisone 1 % cream applied to rash as needed.
- **Cyanocobalamin (vitamin B12) replacement** – 1,000 µg IM or oral weekly ×4 weeks, then monthly.
- **Vitamin D3 supplementation** – 2,000–4,000 IU daily after re■checking levels.
- **Diabetes management** – Initiate metformin 500 mg BID if no contraindications, plus lifestyle counseling.

> **Medication Disclaimer:** A qualified human doctor must make the final prescribing decision and tailor therapy to the individual patient's full clinical picture.

Information Gaps

- Detailed morphology and distribution of the rash (maculopapular vs. vesicular vs. urticarial).
- Presence and severity of respiratory symptoms (cough, dyspnea, chest pain).
- Recent travel, sick contacts, or exposure to infectious agents.
- Pregnancy status (relevant for medication safety).
- Full neurological examination findings (neck stiffness, photophobia, mental status).

Understanding Your Report — In Plain Language

This section explains your diagnostic report in everyday language. It is intended to help you understand what your results mean — **it does not replace the advice of a qualified doctor.**

■ What might be causing your symptoms?

Based on your symptoms, the AI's best assessment is ** **Community■acquired pneumonia with associated fever, headache and an acute pruritic maculopapular rash (possible viral co■infection or drug■related eruption). Underlying uncontrolled diabetes mellitus and vitamin B12 deficiency are contributing comorbidities.** (confidence: ?%). This is a preliminary finding — a doctor still needs to examine you and review any tests before a final diagnosis is made.

■ Things to watch out for (Red Flags)

The following were noticed in your results that need prompt attention:

- SpO_2 = 80 % (significant hypoxia)
- Temperature = 100.4 °F (fever)
- Leukocytosis (WBC = 10,570 cells/µL)
- Potential early sepsis (hypoxia + fever + leukocytosis)

■ Tests you may need

Your doctor may ask you to get the following tests done. Each test helps confirm or rule out possible causes:

- Supplemental oxygen – Titrate to maintain SpO_2 > 94 % (nasal cannula, mask, or high■flow as needed).
- Empiric broad■spectrum antibiotics for community■acquired pneumonia (e.g., amoxicillin■clavulanate or a macrolide) pending cultures.
- Oral antihistamine – Cetirizine 10 mg daily for itch relief.
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■ Possible treatments

These treatments may be considered by your doctor. **Do not take any medication without a doctor's prescription.**

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- Empiric broad■spectrum antibiotics for community■acquired pneumonia (e.g., amoxicillin■clavulanate or a macrolide) pending cultures.
- Oral antihistamine – Cetirizine 10 mg daily for itch relief.

• Topical low■potency corticosteroid – Hydrocortisone 1 % cream applied to rash as needed.

• Cyanocobalamin (vitamin B12) replacement – 1,000 µg IM or oral weekly ×4 weeks, then monthly.

• Vitamin D3 supplementation – 2,000–4,000 IU daily after re■checking levels.

■ Other possibilities the doctor will consider

Medicine isn't always black and white. Here are other conditions that have similar symptoms and may be explored:

• Acute allergic urticaria/atopic dermatitis flare – High IgE and itchy rash; fever & leukocytosis are atypical.

• Viral exanthem (e.g., measles, parvovirus, COVID■19) – Fever, headache, rash, and possible respiratory involvement; COVID■19 can cause hypoxia.

• Meningitis (bacterial or viral) – Headache and fever present, but no neck stiffness, photophobia, or petechial rash.

• Sepsis secondary to occult infection (e.g., urinary tract, skin infection) – Leukocytosis, fever, hypoxia; however, vital signs lack tachycardia/hypotension.

• Drug reaction (e.g., vitamin B complex hypersensitivity) – Temporal relation to vitamin B supplementation; rash is itchy, yet systemic signs suggest infection.

■ What should you do next?

1. Share this report with a qualified doctor as soon as possible. The AI has highlighted areas of concern but cannot replace a physical examination.
2. Get the recommended tests done and bring the results to your appointment.
3. If you experience worsening symptoms — especially difficulty breathing, severe pain, or high fever — go to an emergency room immediately.
4. Do not self-medicate based on this report alone.
